

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO5_AT1 — UNCERTAINTY VISUALIZATION LAB

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO5 – Analysis (BL4)	Assessment:	Assessment Tool 1 – Uncertainty Visualization Lab
Weightage:	20% of CIA	Total Marks:	25 (5 Questions × 5 Marks)
CO5 Statement:	<i>Analyze and construct visualizations that communicate uncertainty using frequency framing, confidence intervals, trend uncertainty bands, HOPs, bootstrapping, and appropriate axis scales.(BL4)</i>		
Student Name:	Kola Amrutha Sandhya	Reg. No.:	192524270
Section / Batch:	AIDS	Date:	27-08-2026

Code of Conduct

I Kola Amrutha Sandhya 192524270 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:Amrutha

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Show all supporting calculations (e.g., confidence interval bounds, residual standard deviation, bootstrap resample means) where relevant.
- Every diagram must include a title, correctly labeled axes, and a legend where relevant.
- Diagrams may be hand-drawn or clearly described in words if drawing tools are unavailable.

Annexure A – UNCERTAINTY VISUALIZATION LAB

Rubrics for Calculation ASSESSMENT RUBRIC
AT1 — Uncertainty Visualization Lab

CO5 · BT4: Analyze ·

This rubric is used to assess student responses to the CO5-AT1 Uncertainty Visualization Lab problems, in which students construct and interpret uncertainty visualizations (frequency framing, confidence intervals, trend bands, HOPs, bootstrapping, and axis scale choice) from given data.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough understanding of the relevant uncertainty-visualization technique (frequency framing, confidence intervals, trend bands, HOPs, bootstrapping, or axis scale choice) and correctly applies it to the given data.	Good understanding of the relevant technique, with only minor gaps in application.	Basic understanding, but with some misconceptions about the correct technique or how to apply it.	Poor understanding of core concepts; applies a technique fundamentally unsuited to the data.
Explanation	25%	Provides detailed, well-reasoned analysis explicitly tied to the specific values in the dataset, including all required calculations.	Provides clear analysis with adequate reasoning tied to the data.	Analysis is present but generic or only loosely connected to the specific dataset given.	Little or no analytical reasoning provided; answers restate the data without interpretation.
Application	20%	Effectively applies construction and	Applies techniques to	Limited application; the	Fails to apply appropriate

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
		calculation techniques to produce a complete, accurate uncertainty visualization correctly matched to the data.	produce a correct visualization, with only minor errors.	visualization partially represents the uncertainty but has notable errors.	techniques; the visualization does not correctly represent the uncertainty.
Organization	15%	The response is clearly structured, with construction and analysis presented in a logical sequence with coherent overall flow.	The response is organized, with only minor issues in flow or clarity.	Basic structure; construction and analysis are present but the response lacks clarity or sequence.	Poorly organized; the response is incoherent or key parts are left unaddressed.
Accuracy	10%	All parts of the answer — construction, calculations, and analysis — are completely accurate and well-suited to the given data.	Mostly accurate, with only minor omissions or a small error.	Partially correct; some elements are missing, mismatched to the data, or incorrect.	Largely incorrect or inappropriate, with major gaps across multiple parts.

Scoring Guide

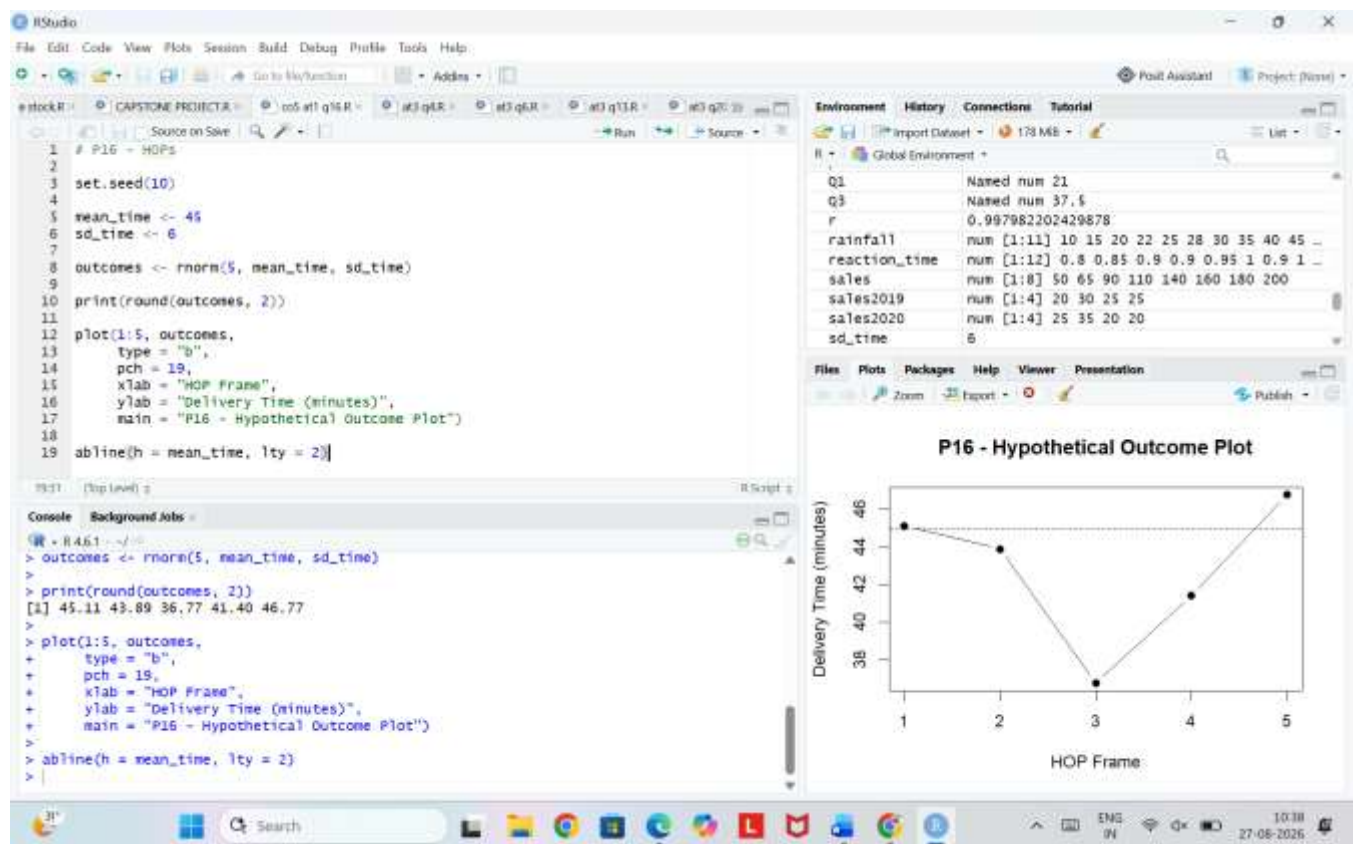
For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Problem 16

Data: A delivery time estimate has a true mean of 45 minutes and a standard deviation of 6 minutes.

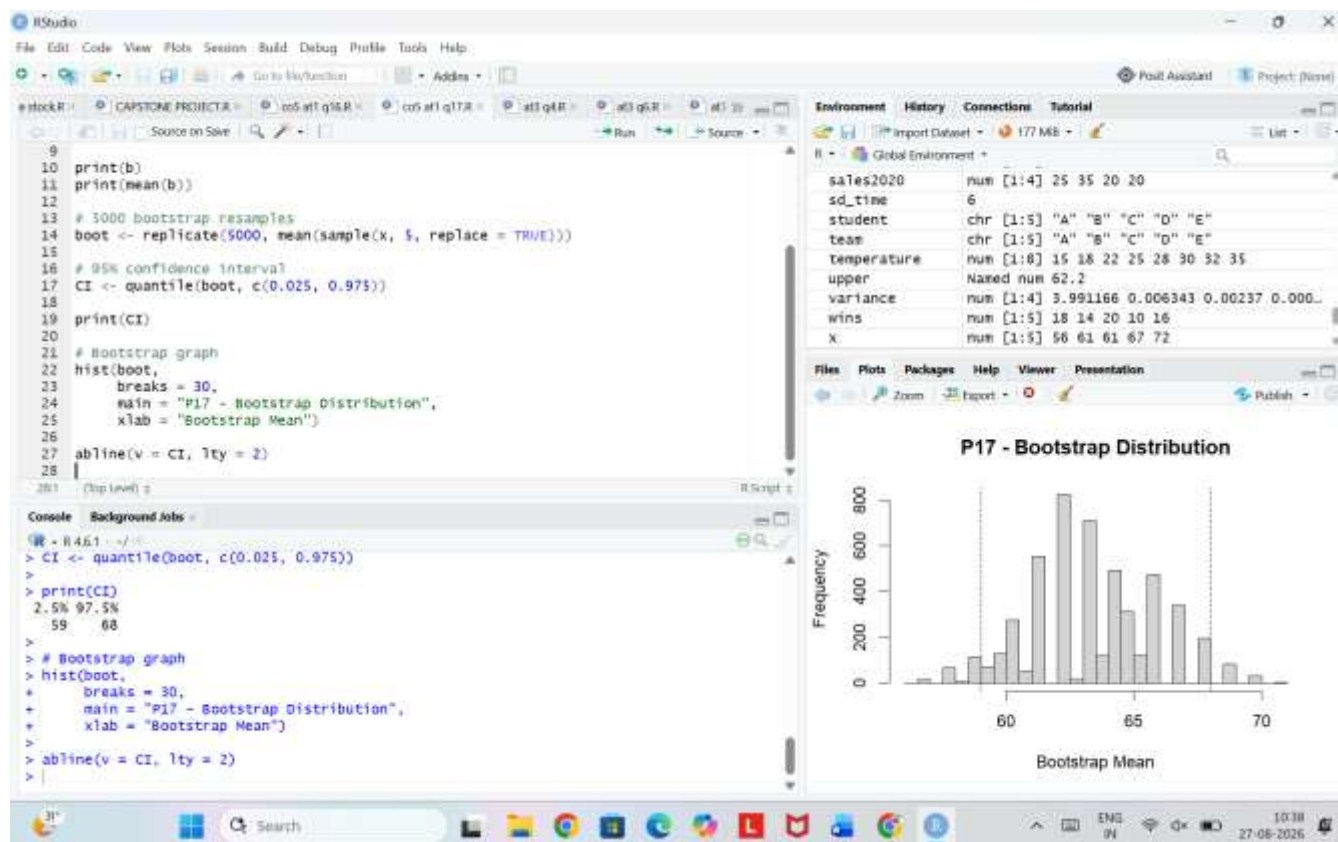
Task: List 5 plausible individual draws that could each form one frame of a Hypothetical Outcome Plot, and explain what a viewer learns from watching all 5 frames in sequence versus a single error bar.



Problem 17

Data: A small sample of 5 product ratings (out of 100): 56, 61, 61, 67, 72.

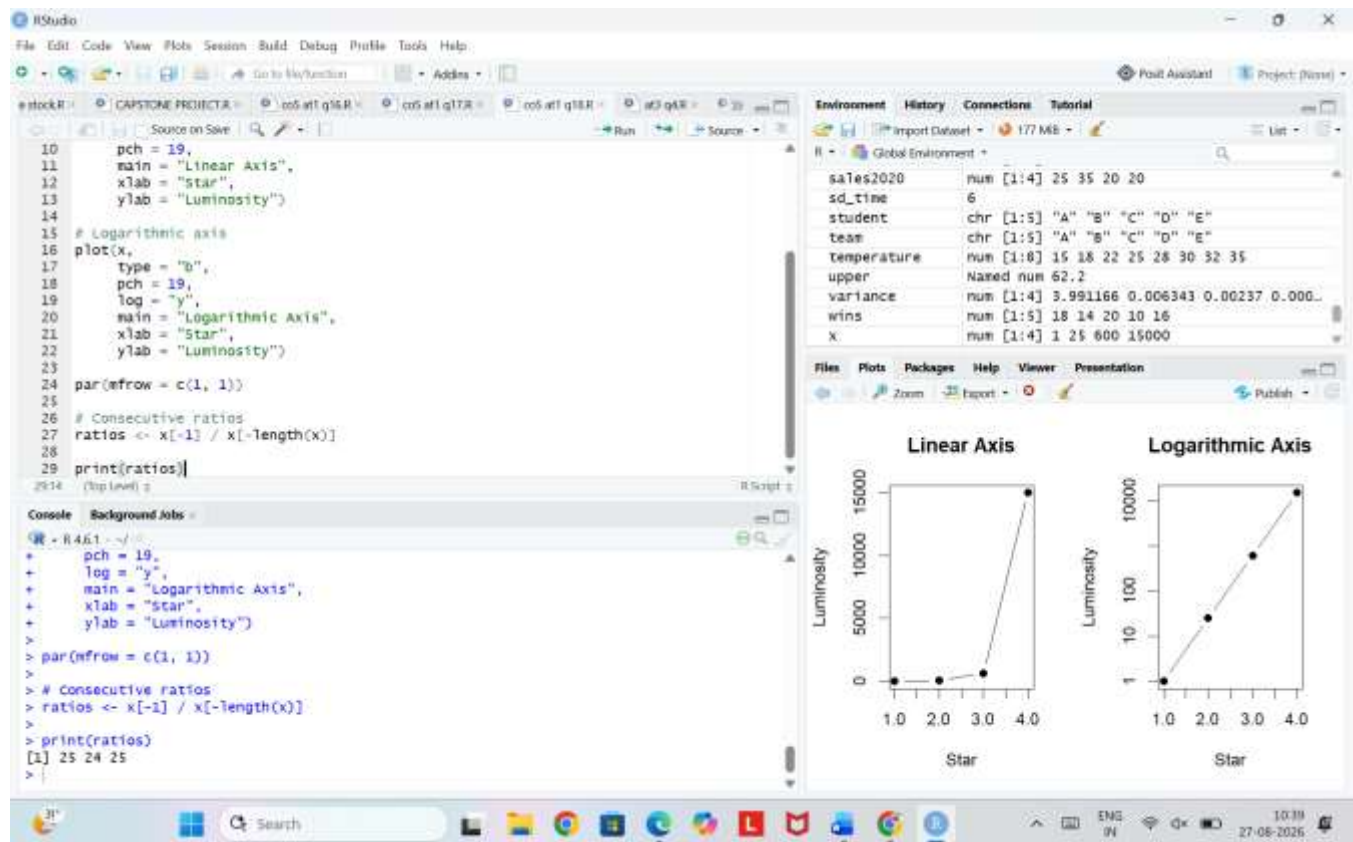
Task: Perform one bootstrap resample (draw 5 values with replacement) and calculate its mean, then explain how repeating this process thousands of times and taking the middle 95% of resulting means gives a confidence interval.



Problem 18

Data: Star luminosities for 4 entities: 1, 25, 600, 15000 solar units.

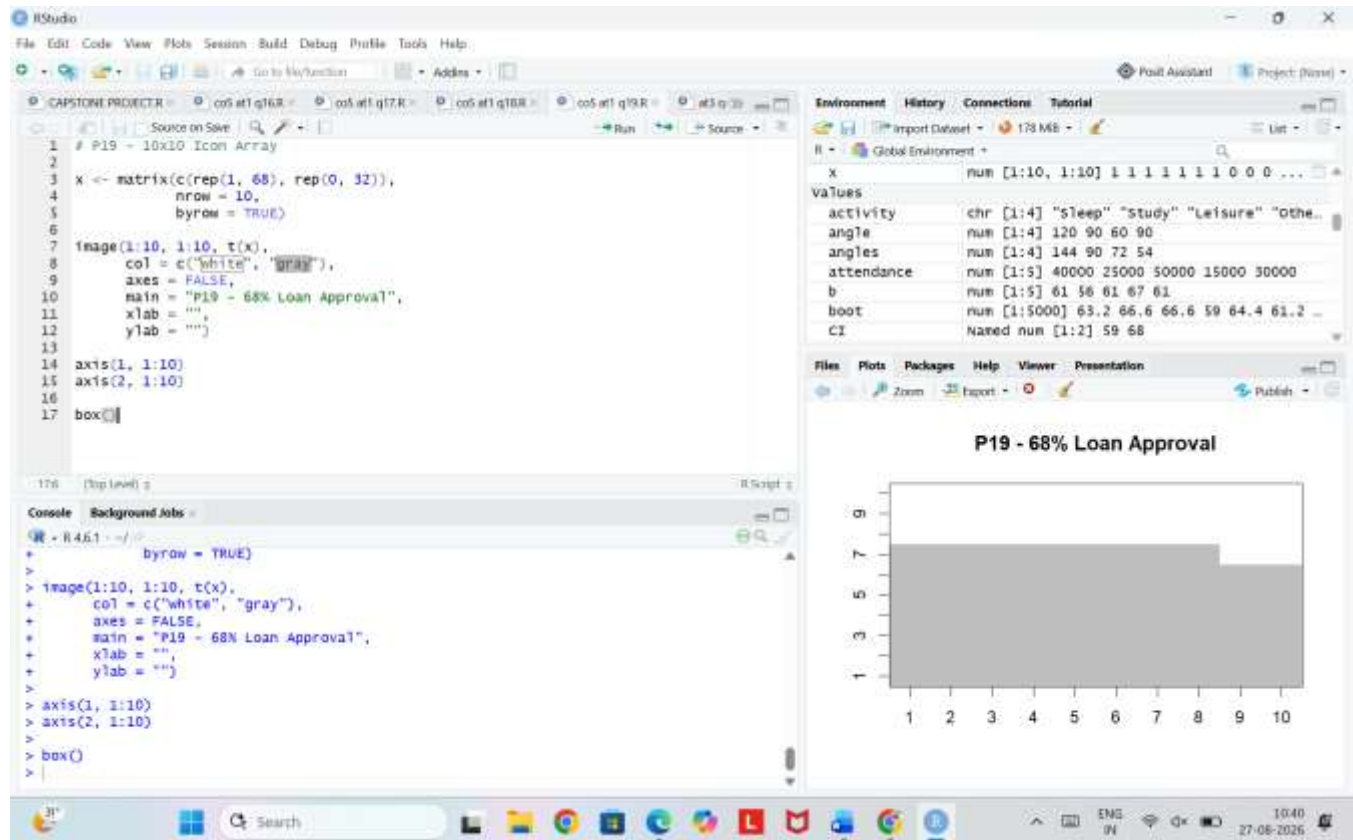
Task: Plot this data on both a linear and a logarithmic axis, calculate the ratio between each consecutive pair of values, and explain which axis reveals the data's structure more clearly, and why.



Problem 19

Data: A report states: “There is a 68% chance of a loan application being approved.”

Task: Reframe this probability as a natural frequency statement, and describe how you would build a 10×10 icon array to represent it.



Problem 20

Data: Candidate 1 vs. Candidate 2 support: $70\% \pm 5\%$ vs. $65\% \pm 5\%$ (95% confidence interval).

Task: Calculate both confidence intervals, determine whether the two overlap, and state whether the higher figure represents a statistically meaningful lead.

